Non Graded Lab Assignment 1 – CSD 207 OOP in Java

For this assignment, you will solve problems based on what you have learned in Chapter: Loops.

Instructions

- Review the chapters from the book and class notes.
- There are 6 questions in this assignment.
- Assignment submitted after due date will not be evaluated.

Due Date and Time: August 17, 2018, in class.

Submitting this Assignment

You do not have to submit this assignment. When you have completed solving all questions, show them to the Instructor.

Grading Criteria

There are no marks to this assignment.

Note: Use class JOptionPane to take input/output wherever required.

Questions:

1. **Write a program that displays the first 50 prime numbers in five lines, each of which contains 10 numbers.**
   An integer greater than 1 is *prime* if its only positive divisor is 1 or itself. For example, 2, 3, 5, and 7 are prime numbers, but 4, 6, 8, and 9 are not.

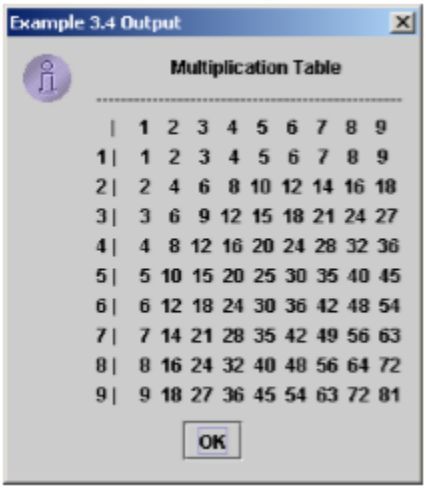
2. **Suppose that the tuition fees for a university is $10,000 this year and tuition increases 7% every year. In how many years will the tuition be doubled?**

3. **Write a program that reads an unspecified number of integers, determines how many positive and negative values have been read, and computes the total and average of the input values (not counting zeros). Your program ends with the input 0. Display the average as a floating-point number. Here is a sample run:**

   Enter an integer, the input ends if it is 0: 1 2 -1 3 0
   The number of positives is 3
   The number of negatives is 1

The total is 5.0
The average is 1.25

4. **Write a program that uses nested for loops to print a multiplication table**



5. **Write a program that sums a series that starts with 0.01 and ends with 1.0. The numbers in the series will increment by 0.01, as follows 0.01 + 0.02 + 0.03 and so on.**

6. **Write a program that prompts the user to enter the year and first day of the year, and displays the calendar table for the year on the console. For example, if the user entered the year 2005 and 7 for Saturday, January 1, 2005 your program should display the calendar for each month in the year, as follows:**

Sample:

```
                 January 2005
---------------------------
Sun Mon Tue Wed Thu Fri Sat
                         1
  2   3   4   5   6   7   8
  9  10  11  12  13  14  15
 16  17  18  19  20  21  22
 23  24  25  26  27  28  29
 30  31
```